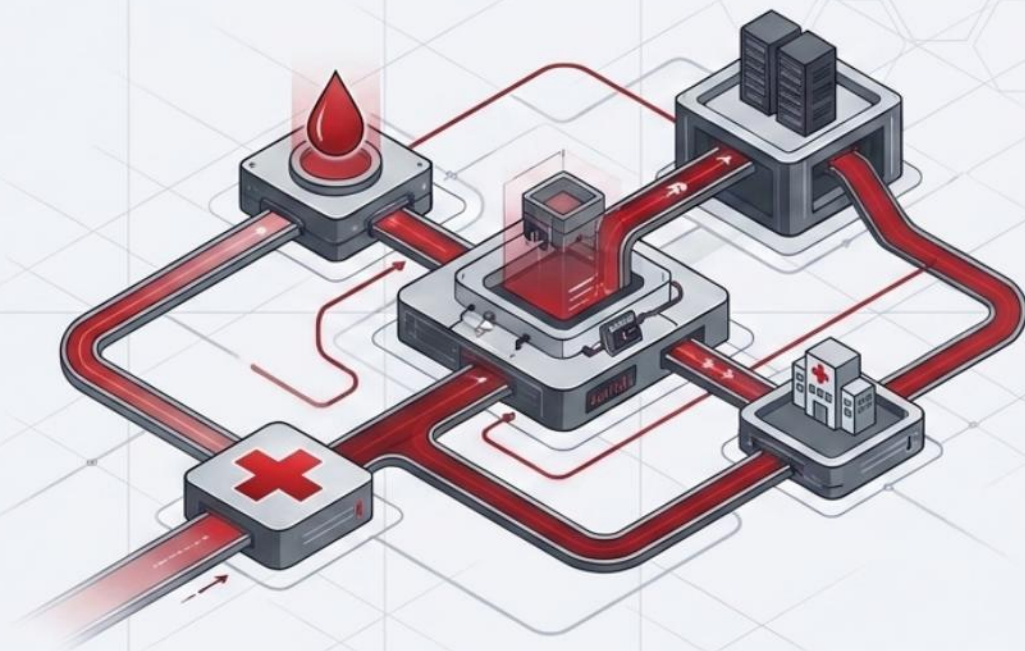


Capitalizing the Biological Supply Chain

Modernizing the infrastructure, logistics, and predictive technology of the American Red Cross blood network.



Sustaining 40% of the Nation's Blood Supply

Volumetric Yield

4.6M
whole blood donations &
1.1M
platelet donations
annually

Donor Base

>2.5M
volunteer donors
across the country

Distribution Footprint

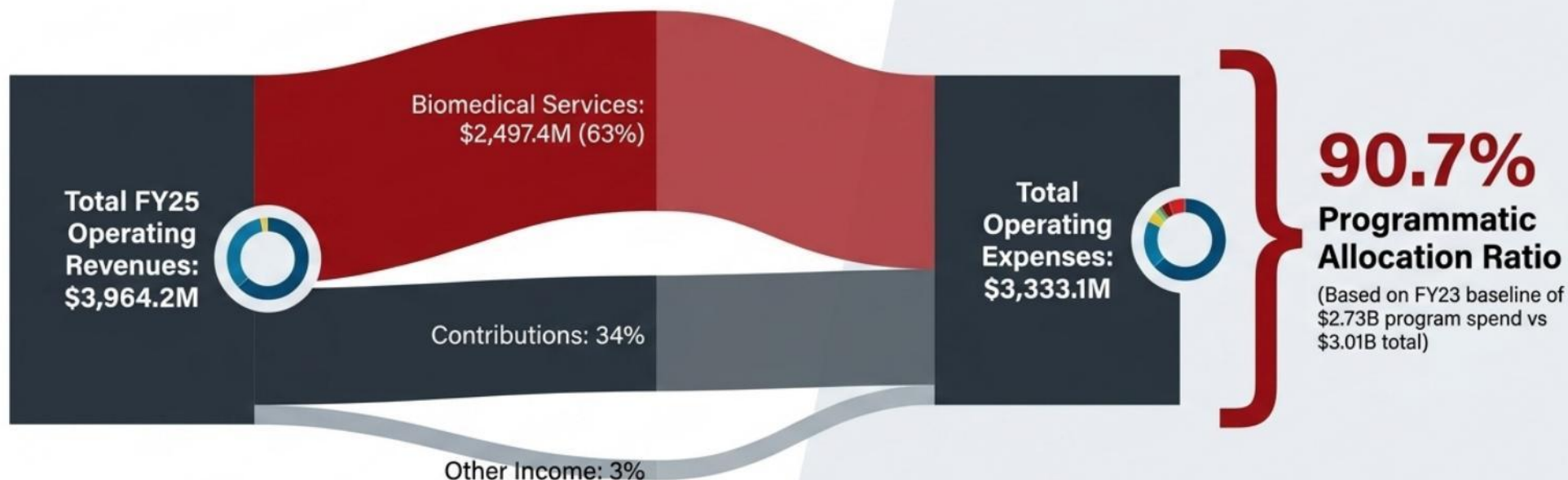
6.4M
blood products
delivered to
~2,500
hospitals and
transfusion centers

Daily Mandate

13,000
whole blood and
3,000
platelet donations required
every single day to meet
clinical demand



The Financial Architecture of a \$3.96 Billion Mission



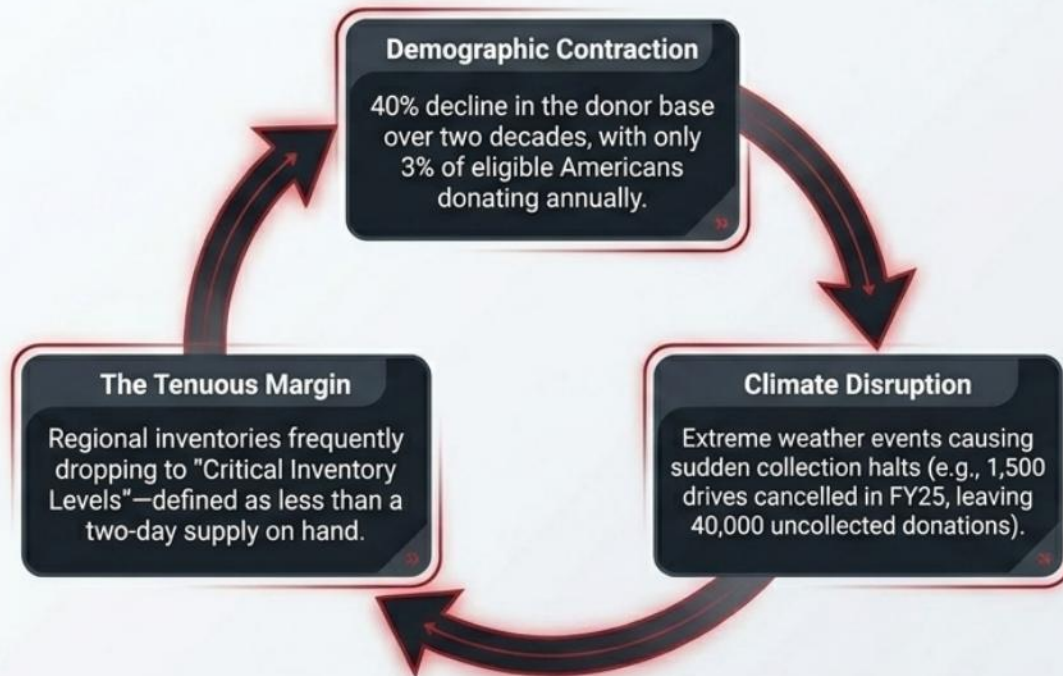
Charity
Navigator
4-Star

Candid
Platinum Seal
of Transparency

BBB
Wise Giving
Alliance

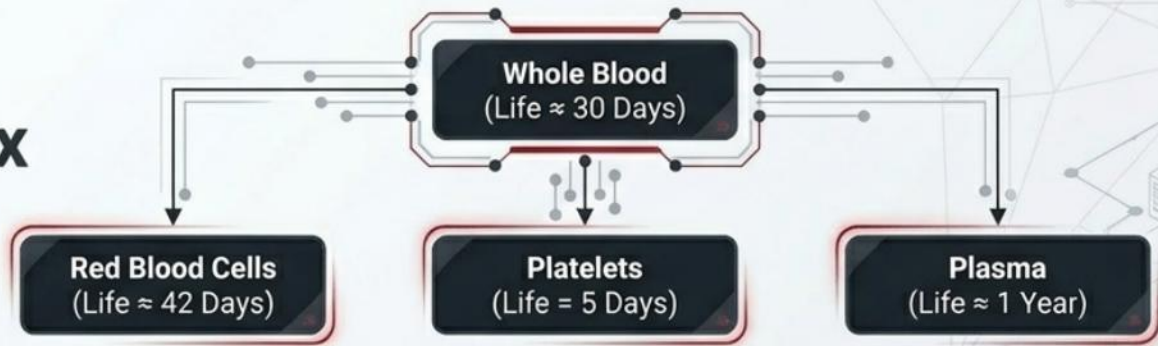


The Fragility of the Traditional Supply Loop



Takeaway: The system depends on a shrinking demographic and is highly vulnerable to unexpected clinical demands (e.g., a single car accident victim can require 100 units).

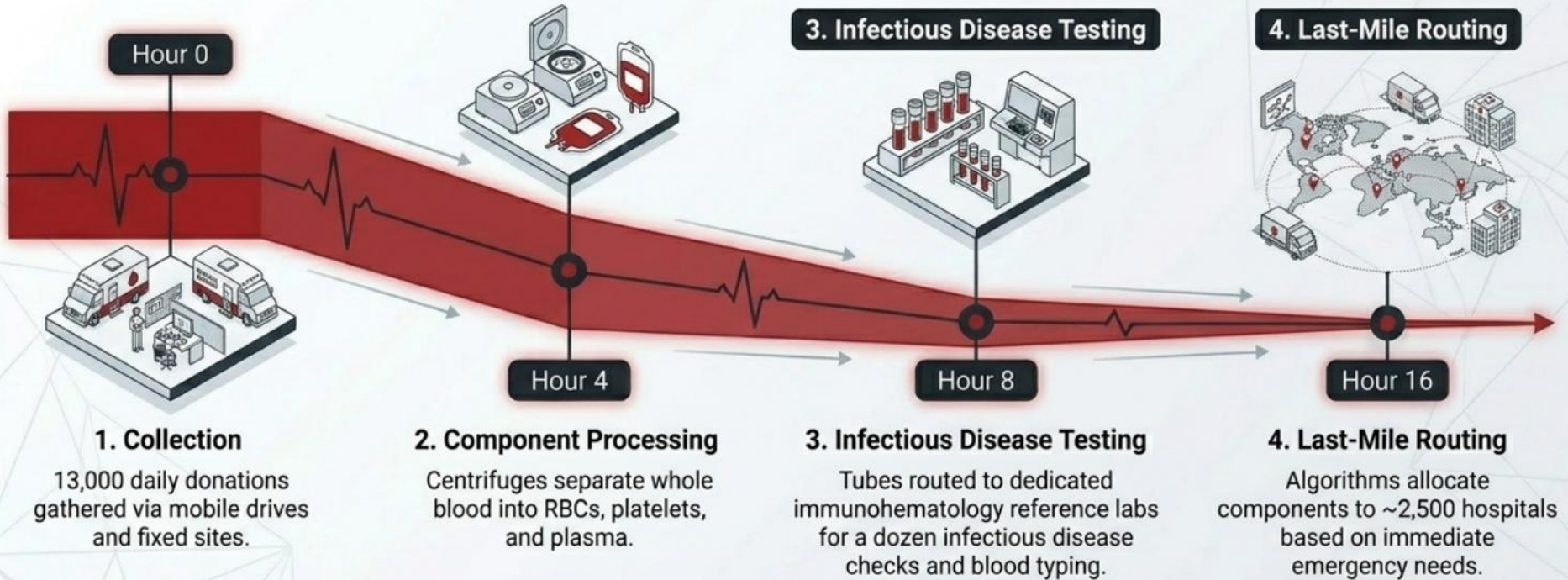
The Biological Constraint Matrix



	Shelf-Life (The Ticking Clock)	Storage Requirement	Primary Clinical Application
Red Blood Cells	42 days	Special Refrigerators	Trauma / Surgery
Platelets	Just 5 days	Room temperature, constant mechanical agitation	Cancer treatment / Transplants
Plasma	1 Year	Freezers	Emergency hemorrhage

Key Insight: Platelets present the ultimate logistical bottleneck. With a strict 5-day expiration and constant agitation requirements, predictive collection and immediate distribution are non-negotiable. »

The Logistics Ticking Clock



Core Message: Every single drop undergoes rigorous manufacturing and screening within hours of collection to beat the strict expiration limits. »

Technology Rewires the Supply Chain

From Reactive to Predictive Operations

Legacy / Reactive

- Paper-based donor tracking
- Retrospective hospital ordering
- Manual collection forecasting
- High vulnerability to regional stockouts



Modernized / Predictive

- AI/ML demand forecasting
- Automated apheresis
- Hyper-local hospital supply tracking
- Autonomous last-mile delivery

Takeaway: Capital investment directly funds the digital infrastructure required to optimize inventory, reduce wastage, and guarantee supply.

Precision Collection: Traditional vs. Automated Apheresis

Manual Whole Blood Collection

- **Process:** Extracts full pint; components separated later in a lab.
- **Time:** ~1 hour total.
- **Yield:** Standard, mixed yield.

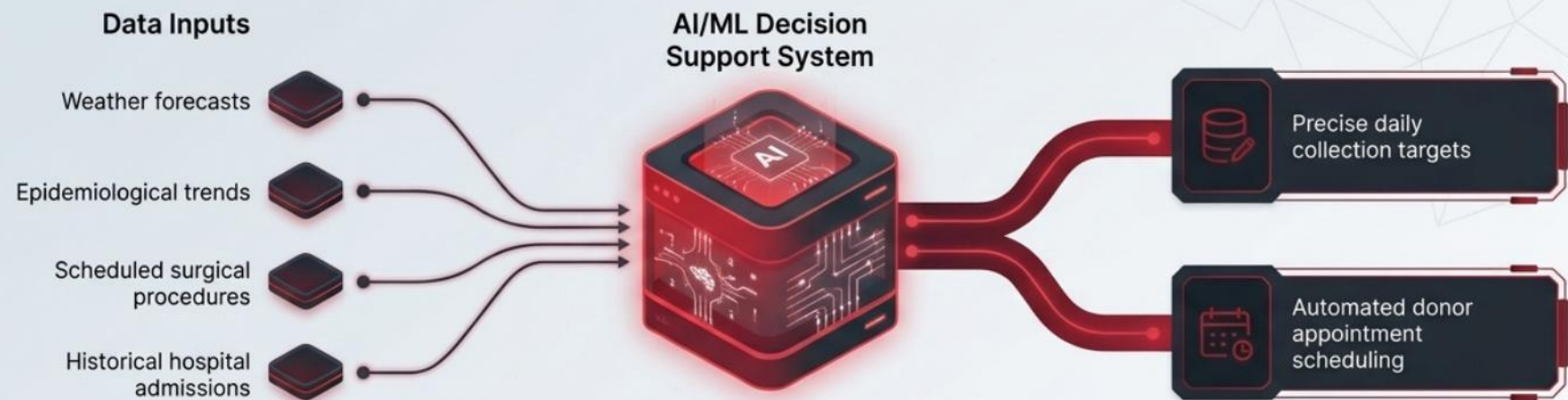
Automated Apheresis (Power Red, AB Elite, Platelets)

- **Process:** Machine centrifuges blood in real-time, **extracts only the targeted component**, and safely returns remaining components and saline to the donor.
- **Time:** 1.25 to 2.5 hours.
- **Yield:** High-concentration, target-specific yield with zero lab-separation delay.



Impact: Maximizes the utility of each donor visit by dynamically collecting exactly what the supply chain demands that day.

AI-Driven Demand Forecasting



The Digital Donor Ecosystem

The Value Exchange

Blood donation now functions as a free preventative health screening, driving higher donor retention.

FY25 Population Health Impact

- 80,000 donors alerted to concerning Hemoglobin A1C levels (prediabetes indicator).
- 69,000 donors notified of Stage 2 hypertension.
- >290,000 sickle cell trait screenings provided to diverse donor populations since 2021.



Autonomous Last-Mile Logistics



The Zipline Paradigm

51%

reduction in maternal mortality from postpartum hemorrhage.

63%

reduction in hospital on-site inventory requirements.

40%

reduction in blood product waste due to rapid, on-demand fulfillment.

Digital Tracking:

The 'Red Cross Delivers' app enables scheduled and urgent dispatch tracking while maintaining strict, real-time chain-of-custody records.

Climate-Resilient Networks in Action

The Scenario

Hurricane Helene knocks out power and water to the Asheville Blood and Platelet Center, a primary distribution site holding over 150 units of highly perishable, life-saving blood.

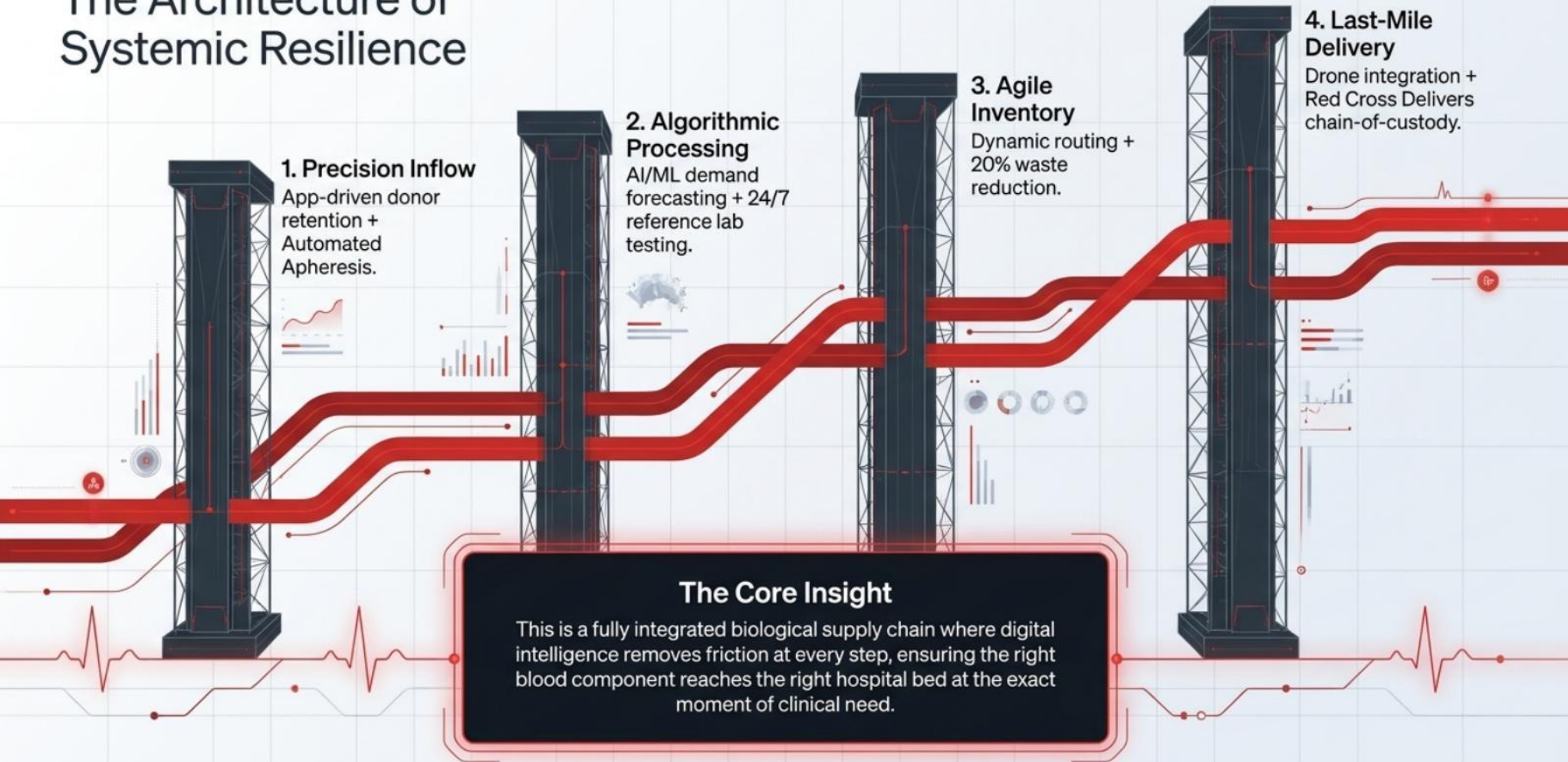
The Response

A specialized Red Cross team navigates four hours through hazardous flood conditions to recover the blood and transfer it to a secure facility, guaranteeing zero loss of life-saving inventory and uninterrupted patient care.



Takeaway: Advanced logistics are meaningless without the human infrastructure and operational resilience to execute under catastrophic conditions.

The Architecture of Systemic Resilience



The ROI of Philanthropic Capital

- **Capitalizing the Future:** Philanthropy does not just buy blood bags; it funds the **predictive algorithms**, automated apheresis machines, and **digital platforms** that modernize national healthcare logistics.
- **Leveraged Impact:** With a 90.7% programmatic allocation and a \$2.5B cost-recovery engine, donor capital acts as pure **innovation funding**, unburdened by administrative bloat.
- **The Ultimate Return:** An unbroken, **climate-resilient lifeline** that guarantees the survival of trauma victims, transplant recipients, and cancer patients nationwide.



Invest in the infrastructure of life.